

Optical Properties of Semiconductor Quantum-Wells

Advanced Lab Course Experiment No. 45
Walter Schottky Institute
Technical University of Munich

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1 Rules for the FoPra

Keep in mind that FoPras are provided by PhD students with very busy schedules. They are interrupting their research to show you some nice physics. Therefore:

1. Be nice.w
2. Be punctual. Yes, even the team member who is otherwise always late!
3. Read the report and be able to answer fundamental questions before the experiment starts.
4. Please name the submitted report: Fopra45_Lastname1_Lastname2_Lastname3.pdf
5. Send the report within 3 weeks after the experiment at the latest. Otherwise you probably have a hard time remembering what you did and the whole learning experience becomes a bit pointless.

2 Introduction

In several areas of application the compound semiconductor Gallium-Arsenide (GaAs) is the better choice than the single element semiconductor Silicon (Si), which is due to different physical properties of these materials. GaAs has a higher mobility of the electrons and a direct fundamental bandgap, to mention only two. Transistors made of GaAs can react much faster than transistors made of Si. In wireless telecommunications (wavelengths in the order of μm and mm) GaAs works up to frequencies which are too high for electrical components made of Si. Furthermore it can emit light directly, a feature in the area of optoelectronics.

Replacing parts of the Ga-atoms by Al-atoms leads to minor changes in the spatial arrangement of the crystal structure. However, the electrical and optical properties are changed drastically. There are several modern techniques to epitaxially grow very thin layers or even alternating layers of different composition and abrupt transitions in the order of magnitude of atomic monolayers. (Epitaxy is the deposition of a crystalline overlayer on a crystalline substrate. The crystal structure of the deposited film equals the crystal structure of the substrate.) Electrical components, such as Quantum-Well lasers or High Electron Mobility Transistors (HEMTs) consist of these layer sequences. The setup of a molecular beam epitaxy system is schematically shown in Fig. 1.

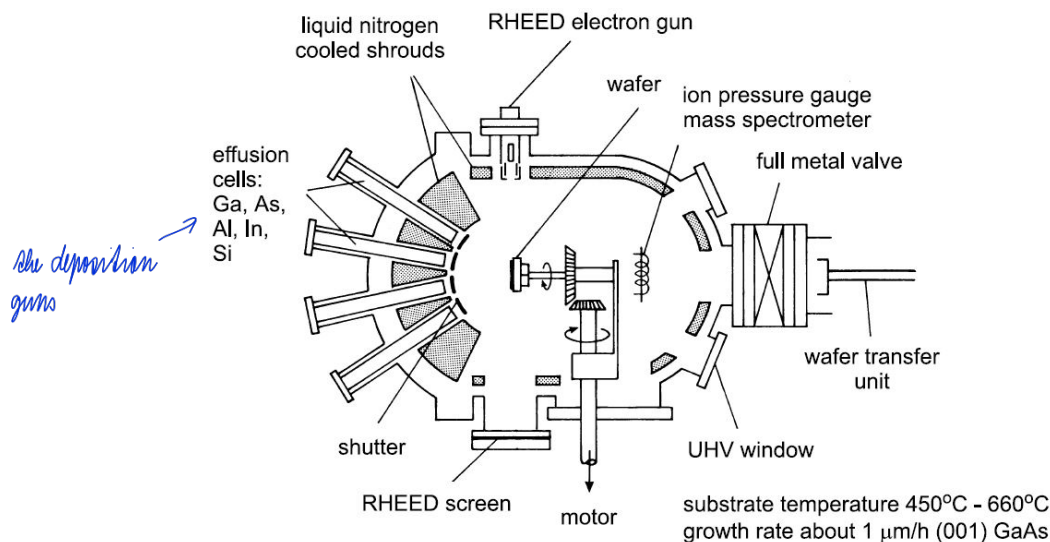


Figure 1: Setup of a molecular beam epitaxy system.

3 Theoretical background

3.1 Calculation of the energy levels in a quantum-well

The energy of electrons in a pure semiconductor can be schematically described by Fig. 2.

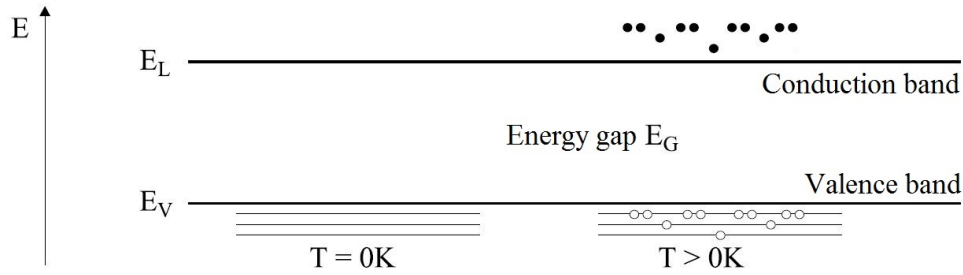


Figure 2: Distribution of electrons and holes for $T = 0 \text{ K}$ and $T > 0 \text{ K}$.

For $T = 0 \text{ K}$ the valence band is fully occupied and the conduction band is empty. They are energetically separated by the band gap E_G . Due to thermal excitation (electrons are excited from the valence band to the conduction band) at finite temperatures ($T > 0 \text{ K}$) some states in the conduction bands are occupied. The missing electrons in the valence band are called holes. *very materials connected*

The epitaxial growth of two different semiconductor materials gives a semiconductor heterostructure. In this structure there are interfaces where two different materials with different band gaps lie directly next to each other. One example: In the material $\text{Al}_x\text{Ga}_{1-x}\text{As}$ the value of the band gap depends on the Aluminum concentration x . The transition can be as small as one atomic layer, consequently the band gap changes over a few Å. This transition can be well approximated to be abrupt, as seen in Fig. 3.

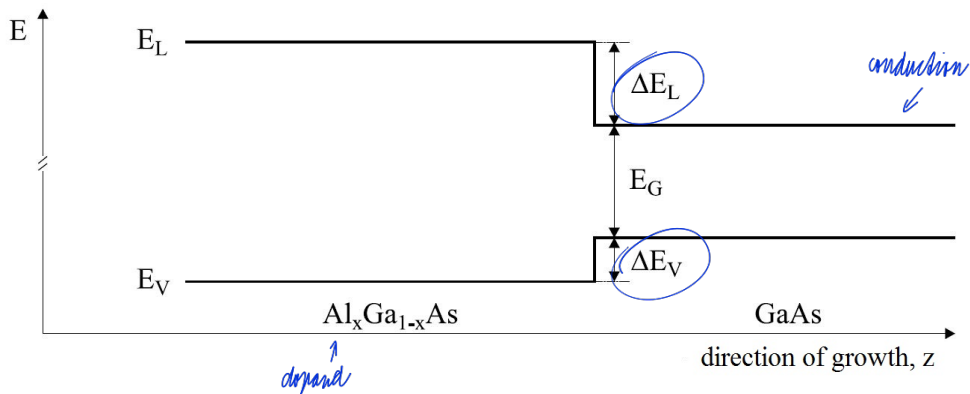


Figure 3: Conduction band and valence band for materials with different energy gaps.

The most important quantities of the heterostructure which depend on the material are the conduction band discontinuity ΔE_L (index L from the German word Leitungsband) and the valence band discontinuity ΔE_V . In the material system $\text{Al}_x\text{Ga}_{1-x}\text{As}$ the ratio $\Delta E_L/\Delta E_V \approx 65/35$ is independent of the Aluminum concentration x .

Adding a mirror image to Fig. 3 gives a so-called quantum-well, as shown in Fig. 4.

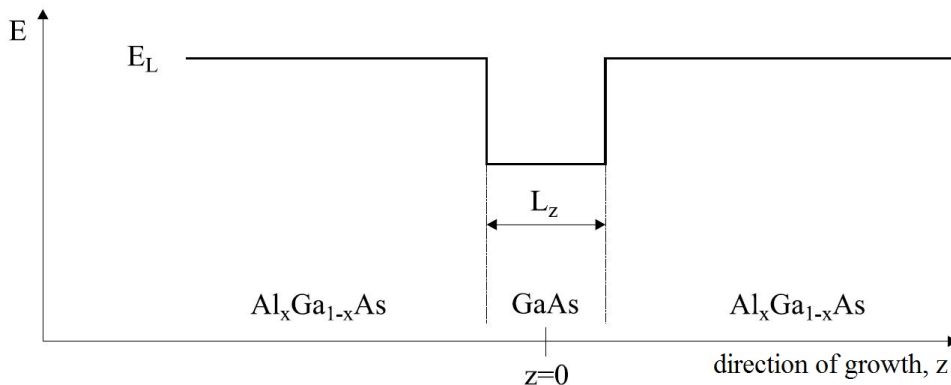


Figure 4: Conduction band of a quantum-well in the material system $\text{GaAs}/\text{Al}_x\text{Ga}_{1-x}\text{As}$.

TO-DO what about valence band?

The states of electrons in the quantum-well - which means they have energies lower than the conduction band energy of $\text{Al}_x\text{Ga}_{1-x}\text{As}$ - must be described quantum mechanically if the width of the quantum-well L_z is in the same order of magnitude as the electron's de-Broglie wavelength. The same is valid for holes in the valence band. The electron in the GaAs layer is therefore "captured" in the direction of growth (z-direction) but can still move parallel to the barrier. For sufficiently thin GaAs layers the energy in z-direction is quantized. A mathematical description of this problem can be done by solving the time-independent Schrödinger equation with a potential only depending on z . In contrary to vacuum, the electron now has a different effective mass for all three directions x , y and z . The eigenvalue equation for the electron in the conduction band is

$$\left[-\frac{\hbar^2}{2} \left(\frac{1}{m_x} \frac{\partial^2}{\partial x^2} + \frac{1}{m_y} \frac{\partial^2}{\partial y^2} + \frac{1}{m_z} \frac{\partial^2}{\partial z^2} \right) + V(z) \right] \psi(\mathbf{r}) = E\psi(\mathbf{r}). \quad (1)$$

Using the ansatz

$$\psi(\mathbf{r}) = \phi_j(z) e^{i\mathbf{k}_{||} \cdot \mathbf{r}}, \quad (2)$$

normal wave along the "free" directions

which includes the free mobility parallel to the layers (wavevector $\mathbf{k}_{||}$), Equation (1) splits into two differential equations:

$$\left[-\frac{\hbar^2}{2m_z} \frac{\partial^2}{\partial z^2} + V(z) \right] \phi_j(z) = \epsilon_j \phi_j(z) \quad (3)$$

and

$$-\frac{\hbar^2}{2} \left(\frac{1}{m_x} \frac{\partial^2}{\partial x^2} + \frac{1}{m_y} \frac{\partial^2}{\partial y^2} \right) e^{i\mathbf{k}_{||} \cdot \mathbf{r}} = E_{||} e^{i\mathbf{k}_{||} \cdot \mathbf{r}}. \quad (4)$$

The kinetic energy of the electron is

$$E = E_{\perp} + E_{||} = \epsilon_j + \frac{\hbar^2 \mathbf{k}_{||}^2}{2m_{||}}. \quad (5)$$

Equation 3 can be solved easily by assuming the electron to move in a box with infinitely high potential wells (particle in a box). The potential $V(z)$ is

$$V(z) = \begin{cases} 0 & \text{for } |z| \leq \frac{L_z}{2} \\ \infty & \text{for } |z| > \frac{L_z}{2} \end{cases} \quad (6)$$

The wave function cannot penetrate infinitely high potential wells. For $z = \pm \frac{L_z}{2}$ the wave function is zero, the wave has a node:

$$\phi_j \left(\pm \frac{L_z}{2} \right) = 0 \quad (7)$$

Due to the fact that the box is symmetric with regard to the middle of the box ($z = 0$) the product $\phi_j \phi_j^*$ must be symmetric with regard to the middle of the box as well. The wave function can therefore only be either symmetric or anti-symmetric.

$$\phi_j(-z) = \pm \phi_j(z) \quad (8)$$

The eigenfunctions of Equation (3) with the boundary conditions (7) are

$$\phi_j(z) = \begin{cases} A \cos \left(\frac{\pi j z}{L_z} \right) & \text{for } j = 1, 3, 5, \dots \\ A \sin \left(\frac{\pi j z}{L_z} \right) & \text{for } j = 2, 4, 6, \dots \end{cases} \quad (9)$$

The kinetic energy ϵ_j can be calculated from Equation (3) and Equation (9) to the following:

$$\epsilon_j = \frac{\hbar^2 \pi^2 j^2}{2m_z L_z^2} \quad \text{for } j = 1, 2, 3, \dots \quad (10)$$

Assuming the potential to have a finite height outside the box gives more realistic solutions. The electron mass inside the box m_t shall furthermore be different from the electron mass in the barrier m_B :

*↑
normal ... ?*

*↑
boundary ... ?*

$$V(z) = \begin{cases} 0 & \text{for } |z| \leq \frac{L_z}{2} \\ V_B & \text{for } |z| > \frac{L_z}{2} \end{cases} \quad (11)$$

$V(z)$ is piecewise constant. Equation (3) therefore has an exact solution. Inside the box $\phi(z)$ is the sum of an incident plane wave $e^{ik_t z}$ and a reflected plane wave $e^{-ik_t z}$ with

$$k_t = \sqrt{\frac{2m_t E}{\hbar^2}}. \quad (12)$$

m_t is the electron mass in the barrier in z-direction. Outside of the box the waves are damped like $\phi(z) \propto e^{-k_B |z|}$ with

$$k_B = \sqrt{\frac{2m_B(V_B - E)}{\hbar^2}}. \quad (13)$$

The wave function inside the barrier is then

$$\phi(z) = \begin{cases} A \cos(k_t z) & \text{even symmetry} \\ A \sin(k_t z) & \text{odd symmetry} \end{cases} \quad \rightarrow |\phi|^2 \text{ has always even symmetry} \quad (14)$$

The probability $\phi\phi^*$ to find a particle very far from the box vanishes. This is due to the boundary conditions. The corresponding wave functions are

$$\phi(z) = \begin{cases} B e^{-k_B(z - \frac{L_z}{2})} & \text{for } |z| > \frac{L_z}{2} \\ C e^{k_B(z + \frac{L_z}{2})} & \text{for } |z| < -\frac{L_z}{2} \end{cases} \quad (15)$$

At the boundary of the barrier $\phi(z)$ and $\frac{1}{m_z} \frac{\partial \phi(z)}{\partial z}$ must be continuous. Then the probability density and the probability current density are continuous in z-direction. From Equation (14), Equation (15) and these boundary conditions two transcendental equations for the electron's energy E as function of the barrier width L_z arise:

$$\sqrt{\frac{m_B E}{m_t(V_B - E)}} \tan\left(\sqrt{\frac{2m_t E}{\hbar^2}} \frac{L_z}{2}\right) = 1 \quad \text{for even states} \quad (16)$$

and

$$\sqrt{\frac{m_B E}{m_t(V_B - E)}} \cot\left(\sqrt{\frac{2m_t E}{\hbar^2}} \frac{L_z}{2}\right) = -1 \quad \text{for odd states} \quad (17)$$

From these two equations E can only be found using numerical iteration for a given L_z . m_t is the mass inside the well, m_B is the mass outside at the barrier. It turns out that the lowest energy level always corresponds to an even state. All in all the energy of the particle in the quantum-well can be described as follows: The particle's movement perpendicular to the potential well corresponds to quantised energy values ϵ_j . The particle's movement parallel to the potential well corresponds to a continuous energy spectrum. All states of energy ϵ_j are referred to as the j th subband in the box, as depicted in Fig. 5.

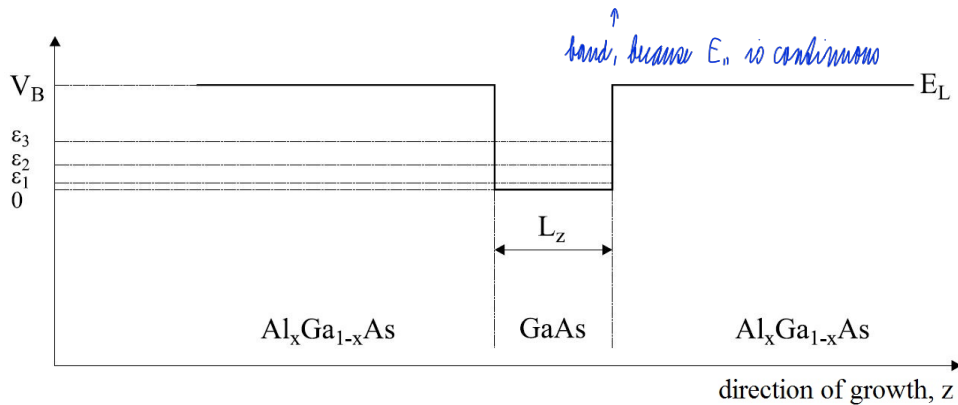


Figure 5: Energy values inside a finite potential well.

In general this calculation holds for both, electrons and holes. Then the corresponding values for the masses m_t and m_B and the potential energy barrier V_B have to be used.

So far Coulomb attraction between holes in the valence band and electrons in the conduction band has not been taken into account. Considering the attractive force between electrons and holes they form a bond which is similar to the bond of an electron and a proton in a hydrogen atom. This formation of electron and hole is called exciton. When calculating the exciton binding energy, one must take into account that the attractive interaction between the charges is shielded by the crystal. Therefore the dielectric constant ϵ needs to be introduced. Additionally the masses of the bonding partners have to be adapted depending on the electron's and hole's size in the crystal. The exciton binding energy is then

$$E_{\text{exciton}} = \frac{m^* e^4}{8\epsilon^2 \epsilon_0^2 \hbar^2} \quad (18)$$

In general this equation only holds for an infinitely extended crystal lattice, the so-called 3D-exciton. Excitons inside of a quantum-well have binding energies which depend on the barrier width, because of the different constraints on the wave functions. In this case the exciton is called the 2D-exciton.

3.2 Optical properties of a quantum-well

When a sample with one or more quantum-wells is irradiated with light, electrons from the valence band are lifted up into the conduction band, under the constraint that the light's energy is high enough, see Figure 6. Depending on the valence band in which the unoccupied state is, the hole is classified as heavy, light or spin-orbit-separated hole. Heavy holes have the lowest energies, followed by the light holes. At low temperatures the electron-hole pairs occupy the energetically lowest state. In semiconductors with direct band gaps most charge carriers recombine emitting a photon. When observing the luminescence in an experiment at low temperatures one consequently sees the radiative recombination of electrons and holes which previously occupied the exciton-ground state in the lowest subband of the corresponding potential barrier.

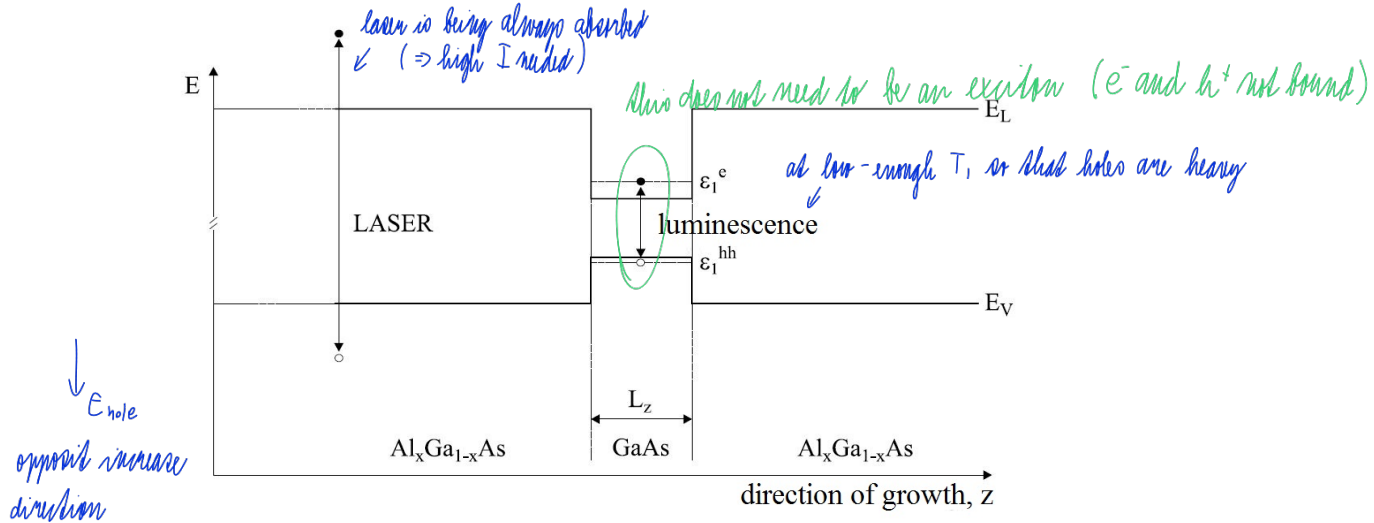


Figure 6: Luminescence in a sample with a quantum-well.

4 Experimental setup

The experimental setup of the luminescence measurement consists of five parts (shown in Fig. 8):

1. Sample and sample holder
2. Imaging optics
3. Light source for excitation
4. Grating spectrometer with control
5. Detector with readout electronics

4.1 Sample and sample holder

The semiconductor sample under investigation consists of a GaAs substrate with a layer sequence of GaAs and $Al_xGa_{1-x}As$ layers grown with MBE (the aluminium concentration x is assumed to be constant in the whole sample). A detailed list of the layers in the sequence is given in the following:

- 2 μm GaAs ... *thick substrate*
- 10 equally wide quantum-wells, separated by 15 nm $Al_xGa_{1-x}As$ each
- 500 nm GaAs
- 200 nm $Al_xGa_{1-x}As$ } *only 2*
- Three quantum-wells with different width, separated by 50 nm $Al_xGa_{1-x}As$ each
- 200 nm $Al_xGa_{1-x}As$ barrier
- 10 nm GaAs cap

The semiconductor pad (3x4 mm²) is mounted on a sample holder right next to the end of the glass fiber (distance to glass fiber $\approx$ 1 mm). Very near to the sample there is also a carbon film resistor for the temperature measurement. The glass fiber not only transmits the excitation light to the sample, but also transmits the luminescence light emitted by the sample to the spectrometer.

4.2 Grating Spectrometer

The setup of a grating spectrometer is schematically depicted in Figure 8. The extended light source under investigation (for example the end of the glass fiber) is imaged on the entrance slit S1 of the spectrometer via diverse optical elements. The entrance slit S1 lies in the focal plane of the spherical mirror M1. M1 reflects a collimated and parallel light bundle which hits a 68mm x 68mm grating (1200 grating elements per 1mm) that is orientated parallel to the entrance slit. The grating reflects the diffracted light into M2 which again focuses the light on the exit slit S2. The spectrometer has selected certain wavelengths from the light passing S1. The light coming from S2 finally hits the detector. Important characteristics of a spectrometer are its inverse aperture ratio (inverse f number) and its spectral resolution.

• Inverse aperture ratio

The aperture ratio is determined by the light trapping of the spectrometer. It is the ratio of the focal length f of the mirror in the spectrometer and the diameter of the grating d . It is a measure for the maximum opening angle for the incident light.

• Spectral resolution of the spectrometer

The spectral resolution $\Delta\lambda$ depends on the minimum distance Δx between two spectral lines which can still be resolved. In the experiment the spectral resolution can be approximately determined by the full width at half maximum of a spectral line of a discharge tube filled with gas at low pressure. In case of a grating spectrometer the spectral resolution depends on the slit width of the entrance and the exit slit. The following formula is a good approximation:

$$\Delta\lambda = \lambda \frac{\sqrt{4 \frac{g^2}{n^2 \lambda^2} - 1}}{2f} \Delta x$$

where:

λ = wavelength

n = order of diffraction

g^{-1} = number of grating elements (= ¹⁵⁰~~1200~~ mm⁻¹)

f = focal length of the grating spectrometer (= ~~32~~ cm)

Δx = slit width (= ²⁵~~250~~ μm)

4.3 Light Source for Excitation

The light source for excitation is a He-Ne gas laser with a wavelength of $\lambda = 632.82$ nm. The reason why a laser is used instead of a broad band light source as for instance a halogen lamp, is the laser's high spectral intensity at a specific wavelength and the laser's directed light.

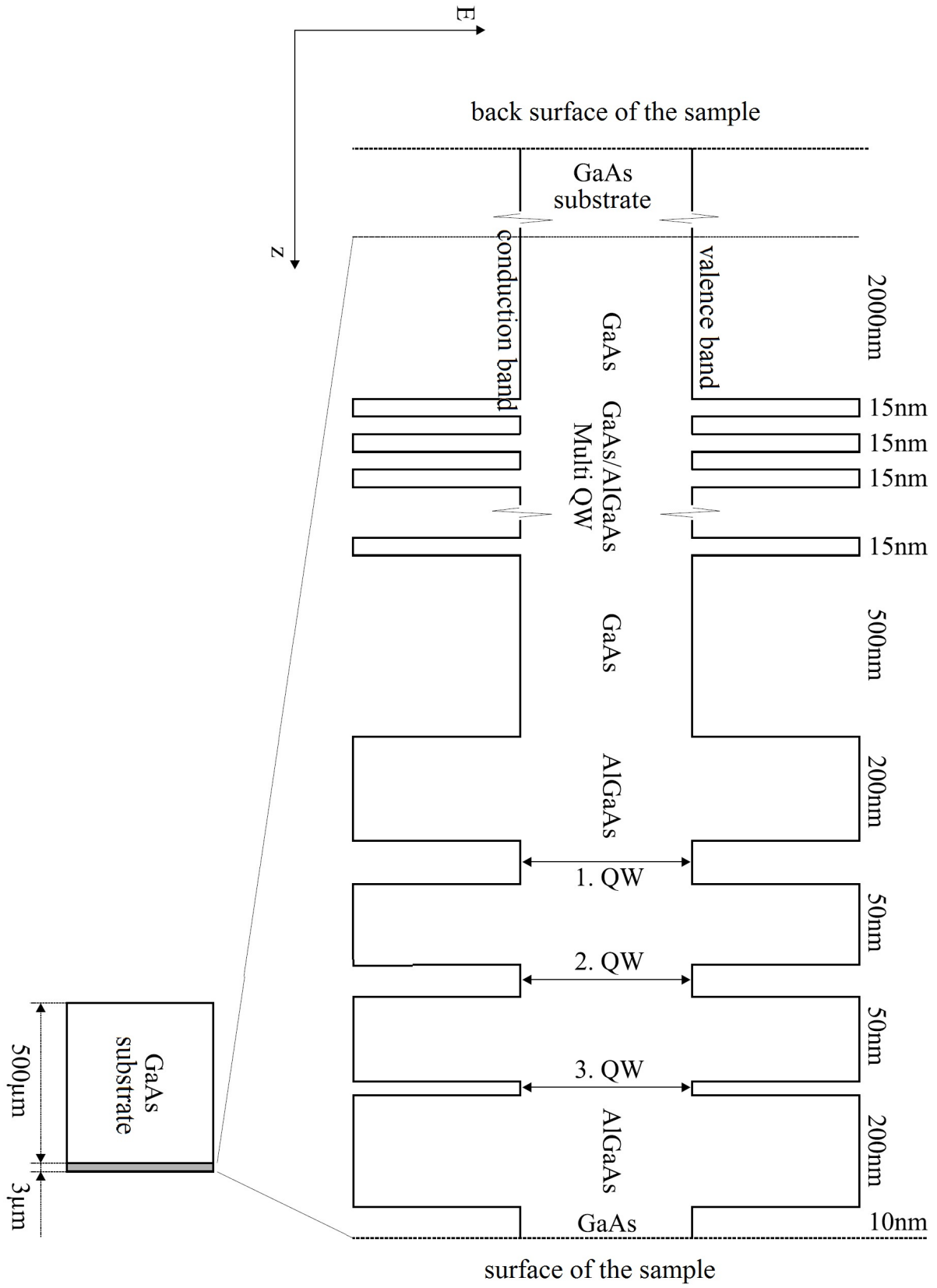


Figure 7: Spatial topology of the conduction band and the valence band of the sample under investigation.

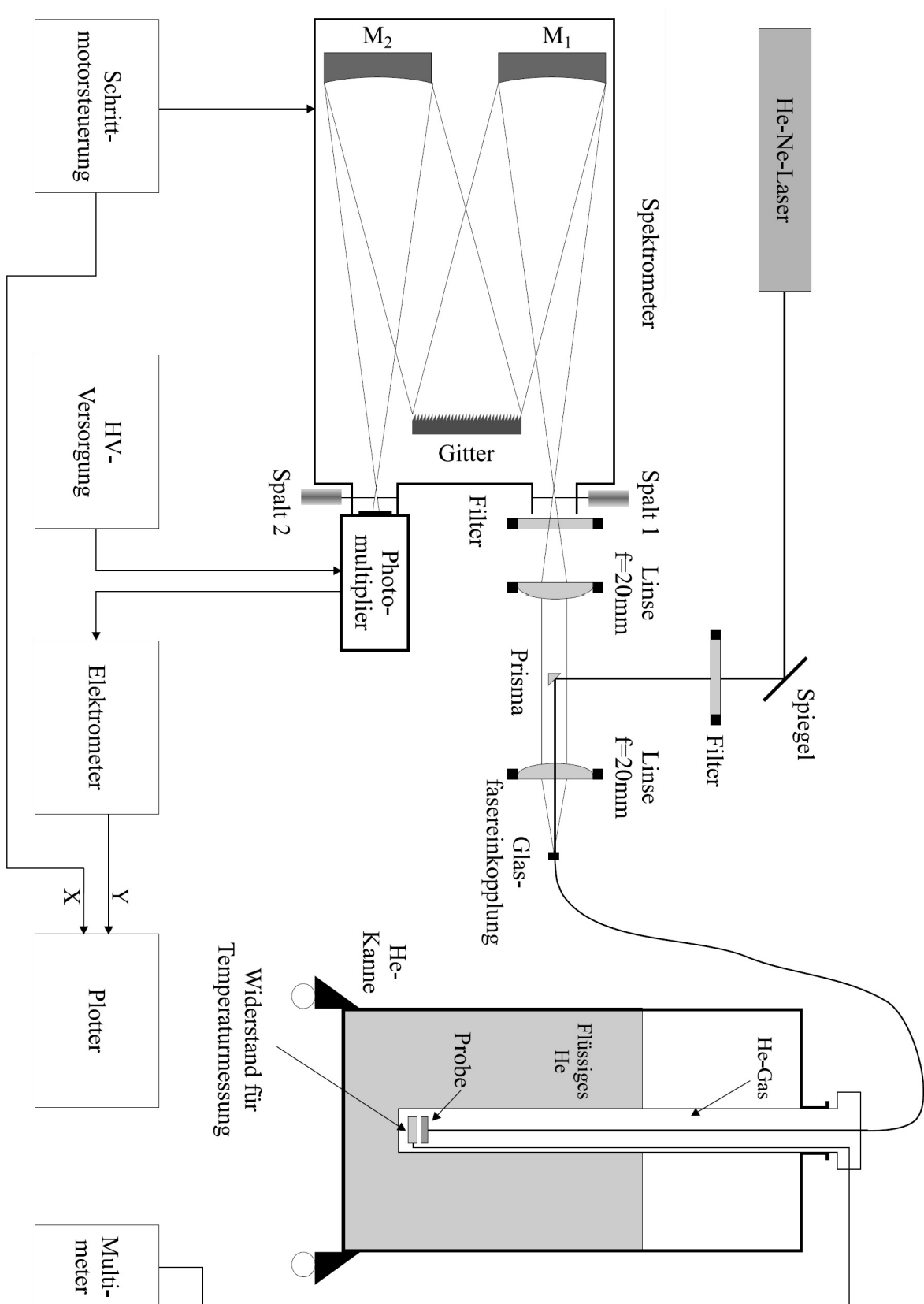


Figure 8: Experimental setup for the photoluminescence measurements.

4.4 Imaging Optics

The setup of the imaging optics is illustrated in Figure 8. The light bundle of the He-Ne laser is focused on the core of the glass fiber by a deflection mirror, a prism and a lens. The fiber transmits the light to the sample on the sample holder which is in good thermal contact to the liquid helium.

Attention! Never directly look into the laser beam! Be careful with reflecting surfaces (mirrors, filter, wrist watch, etc.) near to the laser beam! There are safety glasses at your disposal.

The luminescence light emitted by the specimen in the helium can is transmitted by the same fiber and then focussed by the lenses (photo objective) on the entrance slit S1. The focal length and the distance of the objective to the spectrometer must be chosen in a way that the area on which the light hits the grating is as large as possible. The inverse aperture ratio is about 1:2. In front of the entrance slit one can additionally mount an edge filter to avoid possible laser reflexes inside the spectrometer.

4.5 Detector

Photoemissive detectors, such as e.g. photomultipliers, are based on the outer photoelectric effect. The photo cathode of these detectors is covered by a layer of a material with a small work function. Depending on the spectral sensitivity this material can be an alkali metal or a compound semiconductor like GaAs. Under illumination photoelectrons leave the photo electrode with a kinetic energy that equals the difference between the energy of the infalling photon and the work function. The electrons are accelerated and focussed on the metal surface of the dynode by an externally applied voltage between the cathode and the first dynode. On average, one infalling electron knocks out q electrons at each dynode. This process is repeated at each dynode. Depending on the applied voltage, the dynode's material and the number of dynodes, one photoelectron creates up to several million electrons which then can be registered as electric current.

5 The experiment to be performed

1. Calibrate the spectrometer by using the plasma lines of the He-Ne laser. The corresponding literature values are: 710nm, 730nm, 733nm, 755nm, 760nm, 845nm and 850nm. Please provide a calibration curve in your report.

In the following the calibration does not have to be applied. (we have 12 nm shift)

2. Measure the emission spectrum of the sample at different temperatures. Instead of directly measuring the temperature, calculate its value by using the luminescence of the multi-quantum-well. It is enough to calculate the temperature only for the multi-quantum-well since the temperature is constant on the entire sample.

The following experimental procedure is done for the temperature series:

Cool down:

- Measure the spectrum at room temperature.
- Slowly put the stick with the sample inside of the Helium dewar. The quantum-well lines start to appear.
- Measure several spectra while the sample cools down. The emission lines will shift.
- Measure a spectrum when cold (at 4.2K) and integrate for longer (10s).

Warm up:

- Measure the spectrum in liquid helium (4.2K).
- Slowly pull the rod with the sample out of the Helium dewar. The quantum-well lines start to shift.
- Measure several spectra while the sample warms up and the lines disappear.

For the protocol determine the temperature via the multi-quantum-well's luminescence using the following formula:

$$E_g(T) = E_g(0) - 5.405 \times 10^{-4} \cdot \frac{T^2}{T + 204} \quad (\text{eV})$$

The calculation of the temperature should be done for the cool down procedure. The warm up is a backup measurement in case something went wrong during cool down.

- Use the luminescence measurement in Fig.9 to determine the aluminium concentration x in the $\text{Al}_x\text{Ga}_{1-x}\text{As}$ barrier.

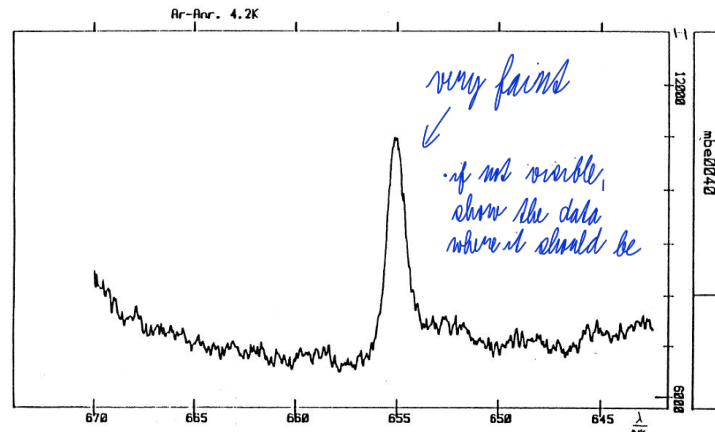


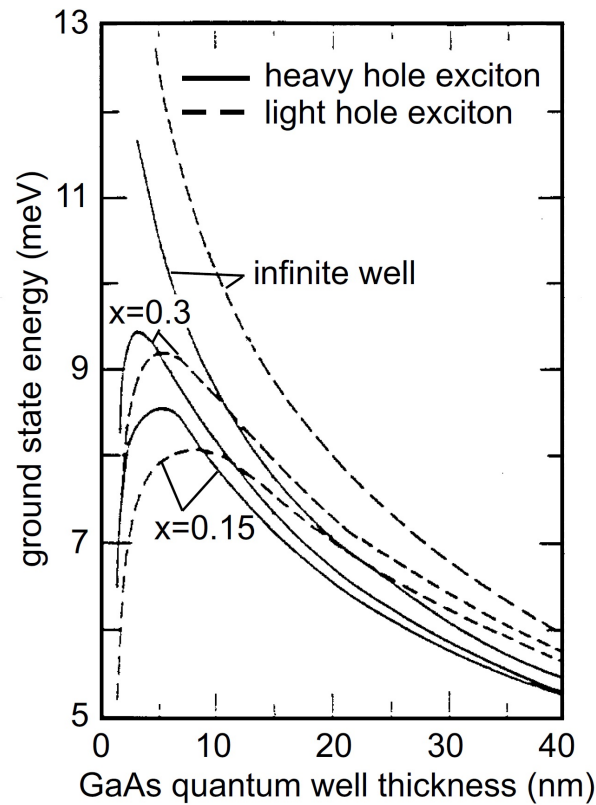
Figure 9: $\text{Al}_x\text{Ga}_{1-x}\text{As}$ luminescence.

- From this determine the changes in potential in the conduction band and in the valence band. Assume that the ratio $\frac{\Delta E_L}{\Delta E_V}$ in the heterostructure is $\frac{65}{35}$.
- Calculate the ground state energy of an electron in a finite potential well as function of the well width with different masses inside and outside the barrier. Solve Equation (16) for L_z and plug in some useful values for E . Do the same for the hole and add the two results graphically. Correct the obtained curve by taking the exciton binding energy into account (Figure 10). Now determine the width of the three quantum-wells and the multi-quantum-well by using the measured energies.
- For comparison determine the ground state energy of the electron and the hole in a box with infinite potential wells. Add the result to the graph.
- From the luminescence energy of the GaAs substrate determine the 3D exciton binding energy and compare it with the theoretical value.
- Compare the measured spectral resolution with the theoretically expected value (Tip: line width).

6 Questions about the experiment

Please answer the following questions in your report. It is enough to write 2-3 sentences to every question.

- Reflect the exact functionality of the grating spectrometer.
- Use the de-Broglie wavelength to estimate how narrow a quantum-well must be in order to have measurable quantum mechanical effects on the electron's energy.
- Why does the binding energy of the exciton depend on the width of the quantum-well?
- Provide a qualitative explanation why the line width of the measured luminescence spectra changes for different widths of the wells.
- Discuss possible effects which could lead to a double line structure of the wider quantum-wells.
- Discuss the changes occurring at the high-temperature measurement.
- Compare the two models (finite and infinite potential well) used for the calculation of the sub band energies and discuss their validity.



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upload plot and retrieve
the curve formula

Figure 10: Binding energy of the ground state of an exciton as function of the box' width for different concentrations x .

7 Literature

1. Ilbach, Lüth: Festkörperphysik: Einführung in die Grundlagen Springer Lehrbuch, 2. Auflage 1988, especially chapter 12.7
2. Kittel: Einführung in die Festkörperphysik Oldenbourg Verlag. 5. Auflage 1980
3. Greiner: Quantenmechanik Teil 1-Einführung Verlag Harry Deutsch, 4. Auflage 1989, Pages 138-149

8 Required numbers

- Energy gap in $\text{Al}_x\text{Ga}_{1-x}\text{As}$ at 4.2K: $E_g = (1.5192 + 1.247x)$ [eV]
- Effective masses:
 - electron $m_e = (0.067 + 0.084x)m_0$
 - heavy hole $m_{hh} = (0.34 + 0.175x)m_0$
 - light hole $m_{lh} = (0.087 + 0.063x)m_0$
- Dielectric constant $\epsilon_{\text{GaAs}} = 12.58$
- Distance of two atomic layers in GaAs: 0.283nm in (001) direction of growth
- Convert energy in wavelength (in air): $E = \frac{hc}{\lambda} = 1239.50 \frac{\text{eV} \cdot \text{nm}}{\lambda}$